

Executive Summary

Our team heuristic evaluation of the budgeting app revealed usability issues in five primary areas: speaking the user's language, minimizing cognitive load, ensuring consistency, providing feedback, and offering shortcuts. In failing to speak the user's language, the app relies on unnecessarily specialized language and uses multiple different terms to refer to the same concept. The app also places excessive burden on a user's memory by providing duplicative functionality with only minor differences in separate subtabs, interfering with a user's ability to develop a coherent mental model about when to use each one. Consistency issues include conflicting placement of form elements, text that is hard to discern whether it is a label or button, as well as stylistic discrepancies in different tabs. The app also provides little visibility into the success or failure of user actions, often providing no confirmation about whether any given action was successful. Finally, the product lacks shortcuts for actions that may be needed to be completed more than once; in particular, there is no way to duplicate entries in some forms where this functionality is likely to be expected, forcing manual entry that is unnecessarily time consuming.

Team Heuristic Evaluation

1. Speak the Users Language / Avoid Jargon

1.1

Issue: The system employs inconsistent and overly complicated terminology, which may lead to user confusion. For example, the "Income" tab has a button called "Ledger", a more niche term for bookkeeping, which encompasses the entirety of the app's purpose, not just income sources. On the "Goals" tab, the tab header reads "Savings & Spending." In addition to being inconsistent with the tab name, as well as the spending side saying "Budget" instead of "Spendings", the header is more jargon than it needs to be, and could be accomplished with a singular term.

Severity: (2) Minor usability issue. While inconsistent and mildly confusing, the naming conventions do not impact overall functionality.

Fixes: For the examples above, "Ledger" could be renamed to "Income" or "Income Sources." Instead of being called "Goals", the tab name and header could be renamed to "Budget", which encompasses the functionality of the tab, and maintains internal consistency within the tab (as it was "Goals"/ "Savings & Spendings" prior).

2. Minimize the user's memory load

2.1

Issue: The application should prioritize recognition over recall to avoid forcing the user to have to memorize many details in order to carry out the desired tasks. For example, within the “Deadlines” tab, there is a sub-tab navigator at the top to switch between “Deadlines” and “To-Do List”. Both contain action items that the user has set out to accomplish. It appears that “To-Do List” is for tasks with no deadlines. If the user wants to get a full view of everything they need to do. They have to constantly switch back and forth between their deadlines and to-do list items, and remember the full scope of each sub-tab's task.

Severity: (3) Major usability issue. Creates the need for the user to have a full mental picture of both tabs at once resulting in unnecessary mental stressors.

Fix: The presence of the ‘Deadlines’ sub-tab accomplishes what the “To-Do List” is meant to, with the addition of having a date assigned to it. Note that deadlines are already not required for the items in “Deadlines” (i.e., I’m able to not select a date and still add an item), so non-rushed tasks can already be added to the same list. Removing the “To-Do List” altogether, as well as the sub-tab navigation bar, they still maintain the functionality (in “Deadlines”), and reduce the need to switch between sub-tabs to see all of your agenda items in both “To Do List” and “Deadlines”.

3. Be consistent, internally and with standards

3.1

Issue: There are inconsistencies between tabs that perform the same function which can lead to confusion. For example, the add functionality is different in every tab. The “Transactions” tab has a blue box at the top with the text “+ New Transaction”. When clicked, users can enter their transaction. The “Income” tab has a “+” next to the text “Add Income” in a blue box, however, when clicked, nothing happens. Additionally, the “Deadlines” tab has two buttons that let you navigate between deadlines and to-do, while the “Goals” tab has one button that toggles between saving and spending. Both have similar functionality but different page designs. Furthermore, the sorting functionality between “Deadlines” and “Transactions” are not uniform where “Deadlines” has “Earliest first” and “Transactions” has “Most recent last”.

Severity (3): Major usability issue. Users are more likely to make a mistake while exploring different tabs due to inconsistency. Also the users might wonder why there are differences between similar functionalities.

Fix: Have a uniform way to add a new “item” in each tab. Have a uniform way to toggle between two sections in a tab.

3.2

Issue: There are inconsistencies in color scheme between graphics, background color, and text color. The color scheme between the “Spending by Category” pie chart and the overall color theme of the app is not compatible. The cancel button in the ‘Deadline & To-Do’ page, when you go to edit a deadline, has black text and a blue background, while the editing on the ‘Transaction & History’ page has black text with white background buttons.

Severity: (2) Minor usability issue: The color contrasts can make it hard for the user to read certain text, especially between black and blue.

Fix: Change the color of the chart to match the app’s color palette and make the buttons in each tab a uniform color.

4. Provide Feedback

4.1

Issue: Confirmation of actions and visible feedback across various tabs. When performing add/remove tasks, virtually no confirmation/error messages appear in the process. After a user adds their income, a pop-up appears asking if the user wants to add a portion to their savings. Clicking “yes” makes the pop-up disappear, and no indicator shows that it was added, unless you navigate to the “Goals” tab to check for yourself. In the “Deadlines & To-Do List” tab, clicking “Add” without completing required fields adds nothing to the list, as expected. However, no error message or alert indicates that the user filled out the form incorrectly.

Severity: (3) Major Usability Issue. Actions feel confusing and meaningless. Without feedback, no mental mapping is established, potentially leaving the user frustrated with their understanding of the functionality.

Fixes: While only two examples were mentioned in this entry, there is a severe lack of feedback across the entirety of the app when adding/removing items. It is strongly recommended that pop-ups or confirmation methods be robustly implemented across the app (perhaps even redirecting to show cross-tab interactions).

4.2

Issue: Lack of visibility and text cues through size, color, and descriptors. Various labelings in the app do not make it clear that an action was completed or that there is something wrong and/or needs to be done. For example, within the “Deadlines”

tab, the due date of each item is either bolded in color or not, seemingly to indicate how close the deadline is. The color differences are extremely subtle, and the size of the text makes it difficult to discern the differences between an upcoming and overdue item (especially for someone with a visual impairment such as color blindness). Across all tabs in the app, the text size is too small, making it difficult to see any visual cues or text descriptors. So much so that it virtually feels like those cues don't exist.

Severity: (3) Major usability issue. The font sizes, coloring, and lack of text cues hide functionality and required actions of the user. The user might miss a deadline payment or be unaware of how their budget is impacted by actions within the app, without mental math. Small text also increases strain on the eyes of the user, and improper coloring/text cues assume visual ability, leading to potential error.

Fixes: Increasing the font size across the app as a whole, as well as adding numerous descriptors to places where changes take place or actions are required. For example, on "Deadlines", additional alerts (perhaps pop-ups) should be displayed for upcoming or overdue items. In addition, text should be included alongside dates, perhaps along the lines of "Due in X Days."

5. Provide Shortcuts

Issue: In the Income tab, there is no way to duplicate lump sum income. This could be useful for folks who complete tasks once in a while that generate revenue, but not on a consistent basis, e.g. mowing a neighbor's lawn, donating plasma, etc. Similarly, in the Deadlines tab, it can be inconvenient for someone who has consistent deadlines on a monthly basis, e.g. paying rent or making car payments, to have to manually add it each time.

Severity: (2) Minor usability issue. While these inconveniences do not interfere with the application's overall functionality, they result in actions taking longer than necessary to complete.

Fixes: In any tab that allows data entry, provide a duplicate button. This button should auto-populate the submission form but not automatically submit it, which saves the user time in re-entering repetitive details, but still allows them to make adjustments if necessary.